Over The Air Computing
Master Thesis
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***Abstract* – Over The Air Computing (OTA Computing) is an emerging technique that leverages the superposition property of wireless channels to perform computations directly during the transmission process. This approach has the potential to significantly reduce latency and energy consumption in distributed systems, making it particularly suitable for applications in Internet of Things (IoT) networks, edge computing, and real-time data processing. This thesis explores the theoretical foundations of Over-the-Air Computation (OAC), including channel modeling, signal processing techniques, and error analysis. Practical implementations and case studies demonstrating the effectiveness of OTA Computing in various scenarios are also presented.**

Contents

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List of Acronyms and Abbreviations

Introduction

0.a Background and Motivation

The increasing need for efficiency in computation-oriented applications where the interest lies in a function of local information, rather than the information itself [1]. The conventional approach of separating communication and computation often proves suboptimal [1], [2], [3]. The potential of Over-the-Air Computation (OAC) to achieve significantly higher computation rates by harnessing interference through signal superposition in a wireless Multiple Access Channel (MAC) [1].

0.b Problem Statement and Research Objectives

Focusing on designing and validating a practical OAC network layout suitable for implementation in a controllable environment[1]. The specific role of Software Defined Radios (SDRs) and MATLAB/simulation tools for prototype validation and addressing practical limitations such as synchronization and channel impairments [1], [4], [5].

0.c Thesis Contributions and Organization

1 Over-the-Air Computation: Foundations and Techniques

1.a Theoretical Foundations of OAC

- *Signal Superposition in the Multiple Access Channel (MAC)*: The distinct feature of OAC where computation is handled by harnessing interference via simultaneous transmissions [1], [3].
- *Nomographic Functions*: Defining the class of functions computable via OAC, which can be expressed as a post-processed sum of pre-processed sensor readings [1], [2], [3], [6].
- *Examples of Nomographic Functions*: Arithmetic mean, weighted sum, maximum/minimum value, and modulo-2 sum[1], [3], [5].

1.b OAC Schemes and Implementation Strategies

- *Analog vs. Digital OAC*: Reviewing uncoded analog transmission (often optimal for simple Gaussian sensor networks) versus digital schemes designed for modern systems (e.g., using QAM or one-bit quantization)[5], [7].
- *Coherent Schemes (CSI Dependent)*: Discussing methods requiring Channel State Information (CSI), such as Truncated Channel Inversion (TCI) or Zero-Forcing (ZF) coordination, for phase and amplitude alignment[1], [7], [8].
- *Non-Coherent Schemes (CSI Independent)*: Examining techniques robust to synchronization errors, such as those using Orthogonal Signaling (FSK/PPM)[1], [9].

1.c Applications Context (e.g., Distributed Consensus)

Application choice (e.g., Federated Edge Learning (FEEL) for gradient aggregation, or average consensus in (Wireless Sensor Networks (WSNs))

[1], [8], [9], [10], [11].

2 System Model and Practical Challenges in OAC

2.a System Architecture and Channel Model

- *Modeling the Network*: Description of Edge Devices (EDs) transmitting to an Edge Server (ES)/Fusion Center (FC), possibly involving multiple antennas (Multiple-Input Multiple-Output (MIMO))[1], [8], [9], [11], [12].
- *Wireless Channel Model*: Consideration of channel impairments, including fading (e.g., Rayleigh fading), noise (Additive White Gaussian Noise (AWGN)), and path loss[5], [7], [8].

2.b Impact of Practical Impairments on OAC Reliability

2.b.i Synchronization Errors

Time, Frequency, and Phase Offsets (TO, CFO, PO): The detrimental impact of imperfect time-of-arrivals and phase alignment on signal superposition and computation accuracy[1], [5], [12].

```
% This is a comment
x = linspace(0, 2*pi, 100);
y = sin(x) + 0.5 * randn(size(x)); % Numbers like 0.5 will be in Math Font
plot(x, y);
```

2.b.ii Channel State Information (CSI) Imperfection

The need for accurate and fresh Channel State Information (CSI) at transmitters or receivers for many precoding/equalization techniques [1], [5], [9].

2.b.iii Waveform and Filter Design

Addressing time sampling error and intersymbol interference Inter-Symbol Interference (ISI) inherent in digital communication transceivers when applied to OAC [5].

2.c Optimization Strategies to Enhance OAC Performance

- *Power Management*: Strategies to mitigate aggregation errors, such as optimizing power control policies to minimize Mean Squared Error Mean Squared Error (MSE) or maximize convergence speed [1], [5], [8].
- *Transceiver Design*: Joint design of pre-processing functions, precoders/beamformers, and post-processing functions (decoders) to match the channel to the desired function and minimize distortion [1], [3], [9], [12].

3 Experimental Implementation using SDR and MATLAB

3.a Experimental Layout and Network Topology

- *Layout Description*: Presentation of the network structure (e.g., a single-cell OAC scenario or an ad-hoc network) using Software Defined Radio (SDR) to represent Edge Devices (EDs) and a Fusion Center (FC)[1].
- *Hardware and Software Tools*: Specific mention of the SDR and the use of MATLAB for waveform generation, digital signal processing, control loops, and simulation[1], [4], [7], [9].

3.b Design and Implementation of the OAC Protocol

3.b.i Selection of the OAC Scheme for Demonstration

Choice of Target Function (e.g., Arithmetic Mean or Majority Vote): Justification based on practical implementation constraints of SDR[1].

3.b.ii Synchronization Mechanisms

Detailed methodology for achieving coarse block synchronization or mitigating synchronization errors (TO/CFO) in the SDR/MATLAB environment (e.g., based on trigger signals or custom control loops)[1], [7], [9].

3.b.iii Pre-processing and Channel Compensation Implementation

Implementation of the chosen modulation/encoding scheme and any necessary pre-equalization (e.g., Channel Inversion precoding if a coherent scheme is selected, or non-coherent coding if robustness is prioritized)[1], [7].

3.c Measurement and Data Collection Methodology

Procedure for varying network parameters (e.g., number of transmitting devices, transmit SNR)[5].

4 Experimental Results and Discussion

4.a Validation of the Implemented OAC Network

Demonstration of successful signal superposition and computation via the SDR platform[1], [2]. Comparison of achieved computation result against the ideal theoretical value[1], [7].

4.b Performance Evaluation under Impairments

Quantifying the performance metrics, such as Mean Squared Error (MSE) or Computation Error Rate (CER), of the implemented scheme[1], [5]. Analysis of the impact of deliberately introduced synchronization errors or varying channel noise (SNR) on the computation accuracy[1], [5]. Comparison with Separation Schemes (Simulated Baseline); Illustrating the resource savings and computation rate achieved by the OAC approach relative to traditional orthogonal access (e.g., Time-Division Multiple Access (TDMA))[1], [2], [3].

5 Conclusion and Future Work

5.a Summary of Thesis Findings

Recap of the feasibility and performance demonstrated by the SDR-based OAC network.

5.b Future Research Directions

Exploring more complex OAC architectures, such as multi-antenna systems MIMO or using Reconfigurable Intelligent Surfaces (RIS)[5], [11]. Further investigation into standardized OAC protocols for compatibility with modern wireless networks (e.g., LTE/5G protocols)[1], [7].

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